

1: 引言第一句: what is the “first star”? 作为核天体物理外门汉, 不明白这个概念。is it “the first stars after the Big Bang”?

引言的描述感觉有些不大直接, 下面的表述如何?

The ${}^7\text{Be}(p, \gamma){}^8\text{B}$ reaction plays a central role not only in the evaluation of solar neutrino fluxes but also in the evolution of the first stars after the Big Bang. Study of this reaction requires the asymptotic normalization coefficient (ANC) for the single proton in ${}^8\text{B}_{\text{g.s.}} \rightarrow {}^7\text{Be} + p$. By using the charge symmetry relation, we obtain this proton ANC with the single nucleon ANC of ${}^8\text{Li}_{\text{g.s.}} \rightarrow {}^7\text{Li} + n$, which is determined with the distorted wave Born approximation (DWBA) analysis of the ${}^7\text{Li}(d, p){}^8\text{Li}$ reaction. The astrophysical S-factors and reaction rates of the direct capture process in the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ reaction is further deduced at energies of astrophysical relevance. The astrophysical S-factor at zero energy for direct capture, $S_{17}(0)$, is derived to be 20.5 ± 3.3 eV b, which is in good agreement with the latest recommendation value. The contributions of the 1^{+} and 3^{+} resonances to the S-factor and reaction rate are also calculated. This work provides an independent examination of the current results of the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ reaction.

2: 第一节应该指出 ${}^8\text{B} \rightarrow {}^7\text{Be} + p$ 的 ANC 可以通过 ${}^7\text{Li}(d, p){}^8\text{Li}$ 的研究得到, 这样也可以跟第二节有个比较好的衔接。

下面我只评论第二节的内容:

2: 第二节第一句话, 建议改成:

When the ${}^7\text{Li}(d, p){}^8\text{Li}$ reaction is measured at conditions when it is peripheral, such as at sufficiently low incident energies, the single neutron ANC of ${}^8\text{Li} \rightarrow {}^7\text{Li} + n$ can be extracted by matching the theoretical calculations (in this work, we use the DWBA method) to the experimental angular distributions of the differential cross sections:

Eq.(1),

with

Eq.(2),

where ...

3: 第二页第一列倒数第四行: ..., j_f mean the ... $\rightarrow$..., j_f are the ...。下面这一行, 建议去掉 “initial and final nuclei”

4: 第二页第二列第一句: 建议改成: In this work, theoretical calculations of the transfer cross sections are made with the finite-range DWBA using the computer code FRESKO[31]. (FRESKO 不只是一个 finite-range DWBA code, 实际上 Ian 写这个程序正是为了可以做超出 DWBA 的计算)

5: 第二页第二列第二、三句说得不太明确, 看下面说法如何?

Two neutron ANCs, $C_{p_{1/2}}^{({}^8\text{Li})}$ and $C_{p_{3/2}}^{({}^8\text{Li})}$ are needed in the theoretical analysis of the ${}^7\text{Li}(d, p){}^8\text{Li}_{\text{g.s.}}$ reaction, which associates to the two main neutron configurations in the ground state of ${}^8\text{Li}$. These ANCs are obtained by taking their ratio $(C_{p_{1/2}}^{({}^8\text{Li})})^2 / (C_{p_{3/2}}^{({}^8\text{Li})})^2$ fixed to be 0.13(2) [33].

6: 第二页第二列倒数第七行: calculated out $\rightarrow$ calculated

7: 这一句中的 different geometry parameters 到底指的是什么? 是 r_0 , 还是 a 还是两个都变? 图 2 的横坐标为什么是 single-particle ANC? 这个不是不可以, 但是如果这样的话, 需要在文字中做明确的说明, 说 single-particle ANC 是由 r_0 或者 a 完全决定的。不过为什么要搞得这么麻烦呢?

8: 表 2 中, $0.49 \pm 0.083 \rightarrow 0.49 \pm 0.08$, 其它的类似